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A retrieval augmented generation based optimization approach for medical knowledge understanding and reasoning in large language models

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ABSTRACT

Based on the existing Retrieval Augmented Generation (RAG) technology, this study proposes innovative solution to better address the hallucination issues of current large language models. By optimizing data processing, prompt engineering, and multi-retriever fusion, it resolves issues such as semantic capture bias, inaccurate context retrieval, information redundancy, hallucination generation, and length limitations. Data processing focuses on text cleaning, disambiguation, and removing redundant information to enhance consistency. Prompt engineering aids the model in better understanding the task. The adaptive weight fusion of sparse and dense retrievers improves context retrieval accuracy. Experiments conducted on the CCKS-TCMBench dataset for medical knowledge understanding and semantic reasoning show that the optimized model significantly outperforms the baseline across all evaluation metrics.

1. Introduction

In the natural language processing field, large language models (LLMs) such as GPT-4 and Qianwen have achieved significant advancements, enabling them to generate fluent text and complete a wide range of complex tasks. In the healthcare domain, LLMs hold immense potential for assisting in diagnosis and patient communication [1]. Med-PaLM 2, improved and fine-tuned, performs excellently in medical question answer benchmarks and real scenarios, is recognized by physicians [2]. However, these models still face limitations, including hallucinations, outdated information, and data security concerns. Their inherent limitations become particularly pronounced in medical scenarios where accuracy, timeliness, and reliability are paramount [3]. For instance, “hallucinations” generated by LLMs can lead to incorrect medical advice, posing serious threats to patient well-being. Furthermore, the rapid evolution of medical knowledge, including new therapies and drug interactions, means that LLMs training data can quickly become outdated, failing to provide the most current clinical guidance. Consequently, there is an urgent need for a solution that can integrate the powerful language understanding and generation capabilities of LLMs with authoritative, real-time, and secure external medical knowledge bases, thereby ensuring that the output information in healthcare applications is accurate, reliable, and aligned with the latest clinical practices.

To address the current challenges in knowledge understanding and reasoning, some scholars have proposed new methods based on the existing Retrieval-Augmented Generation (RAG) technology. RAG combines the generative strengths of LLMs with the precision of retrieval systems, allowing models to access external knowledge sources during text generation, thereby improving the output quality. Luo et al. [4] proposed a method involving data collection, information extraction, question generation, answer generation, fine-tuning, and evaluation, analyzing LLMs and validating their effectiveness in the medical domain. Xiong et al. [5] introduced the medical information retrieval augmented generation evaluation benchmark, conducting large-scale experiments that showed significantly improved the accuracy of six different LLMs, bringing their performance to GPT-4 levels. Long et al. [6] proposed the Bailicai framework for medical knowledge injection, which improves model performance in healthcare by integrating knowledge boundary identification, directed acyclic graph task decomposition, and medical knowledge injection modules. Kresevic et al. [7] introduced a method using LLMs to improve healthcare by

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converting clinical guidelines into a more manageable format, enhancing LLM understanding through prompt engineering and text reconstruction strategies. Lozano et al. [8] proposed the automatic literature summarization tool Clinfo. ai, which was evaluated using the PubMedRS-200 dataset. This system retrieves relevant medical literature and generates summaries based on user queries, improving the efficiency and quality of information retrieval in healthcare. Jeong et al. [9] introduced a method combining self-supervised learning and reinforcement learning to address knowledge-intensive tasks and open-domain question answering in bio-medicine, helping doctors find relevant information more quickly and efficiency. Amugongo et al. [10] explored the limitations of LLMs in healthcare and how RAG technology can mitigate these issues. They conducted a systematic review of existing literature, analyzing the application, advantages, disadvantages, evaluation frameworks, and ethical considerations of RAG in different stages. Salemi et al. [11] proposed the eRAG method, where downstream LLMs act as document annotators to generate labels, which are then used to assess the performance of retrieval models in RAG systems. Chen et al. [12] introduced the Retrieval Augmented Generation Benchmark, using the latest news articles and search engine documents to evaluate LLM performance on noise robustness, negation rejection, information integration, and counterfactual robustness, providing guidance for addressing challenges in information retrieval. James et al. [13] developed the RAG framework, automatically assessing answer accuracy, question relevance, and context retrieval relevance. It was trained using the WikiEval dataset, and experiments demonstrated a high correlation between predictions and human judgments. Lakatos et al. investigates the performance of Retrieval-Augmented Generation (RAG) and Domain-Specific Fine-Tuning (DFT) in the development of AI-driven knowledge-based systems, finding that RAG consistently outperforms DFT across multiple evaluation metrics [14].

In summary, large language models, due to their extensive parameters and complex structures, require significant resources for fine-tuning or retraining and lack the domain-specific knowledge to respond to questions in dynamic scenarios. RAG addresses these challenges by retrieving relevant external knowledge during the retrieval phase and integrating it into the user's query during the generation phase through structured prompt templates. The model, guided by knowledge-augmented prompts, leverages vast parameters to generate more accurate answers. This approach not only improves the accuracy and reliability of the model's responses but also reduces resource costs, providing strong support for the application of LLMs in healthcare fields.

2. Related work

2.1. Large language models

Large language models [15] have received much attention in recent years due to their huge number of participants and powerful language comprehension and generation capabilities. Their training process usually consists of three stages: pre-training [16], supervised fine-tuning [17] and reinforcement learning [18]. Pre-training is performed on large-scale unlabeled data to understand linguistic structures and semantic patterns. Supervised fine-tuning uses manually labeled datasets to optimize the model for a specific task. Reinforcement learning further improves the quality of generation, rewarding or penalizing the generated results and improving the decision-making ability of the model. Through these steps, the big language model learns rich semantic knowledge and performs well in a wide range of tasks. New technologies in artificial intelligence have driven the development of Medical Large Language Model (MLLM) [19]. Due to the complexity and high stakes of medicine, where accuracy and reliability are crucial in tasks such as diagnosis, decision support, and knowledge answering. MLLMs require general-purpose language capabilities as well as specialized medical knowledge and reasoning. Models are pre-trained on large-scale medical data (e.g., clinical records, electronic medical records, medical literature, imaging data, etc.) to learn healthcare-specific language patterns and pathological relationships, and subsequently supervised fine-tuning is performed using annotated medical data (e.g., disease diagnosis, drug recommendations) to improve performance. In recent years, significant progress has been made in medical applications based on big language models. OPENAI and Microsoft's GPT series are used for text generation to assist in medical literature writing and records organization [20]. Google's MedPaLM model, which is trained specifically for healthcare, demonstrates strong medical question and answer capabilities [21]. IBM Watson Health applies big language models to clinical decision support systems to help physicians make more accurate diagnostic and treatment decisions [22]. In the clinical decision support system, MLLM combines with electronic medical record data to provide diagnostic suggestions, drug recommendations, and condition monitoring in real time, dramatically improving medical efficiency and accuracy [23].

Although medical large language model has made significant progress in the fields of clinical diagnosis and health management, it still faces challenges in practical applications. Firstly, the sensitivity of medical data requires models to be used in strict compliance with privacy protection regulations. Secondly, the diagnoses or recommendations generated by models may lack reliability when dealing with rare diseases or complex cases. These issues limit their application in clinical settings and further optimization is urgently needed. To address these challenges, RAG technology is considered as a potential solution that combines the retrieval system with the generative model, which can effectively alleviate the data privacy and enhance the reliability and control ability of the model.

2.2. Retrieval augmented generation technology

Large language model optimization focuses on cue word engineering RAG and efficient fine-tuning, the generic workflow of RAG is shown in Fig. 1, and the model fine-tuning process is shown in Fig. 2. The comparison between RAG and model fine-tuning is shown in Table 1. RAG is a technique that combines retrieval and generation to enhance the quality and accuracy of large language model generation by exploiting external data sources [24]. Firstly, the RAG receives user input (e.g., questions or topics) and retrieves relevant text snippets as context from data sources, which include web pages, documents, or databases. Next, the system merges the user input with the retrieved context and adds guidance information (prompt) [25], which is submitted to a large language model (e.g., GPT) to generate an answer or summary. Finally, RAG extracts or formats the required information from the generated

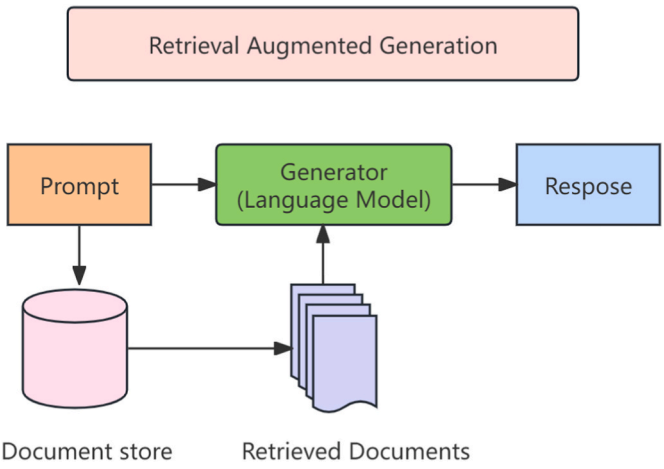


Fig. 1. Retrieval augmented generation workflow overview: prompt processing, relevant document retrieval from a document store, and response generation by a language model.

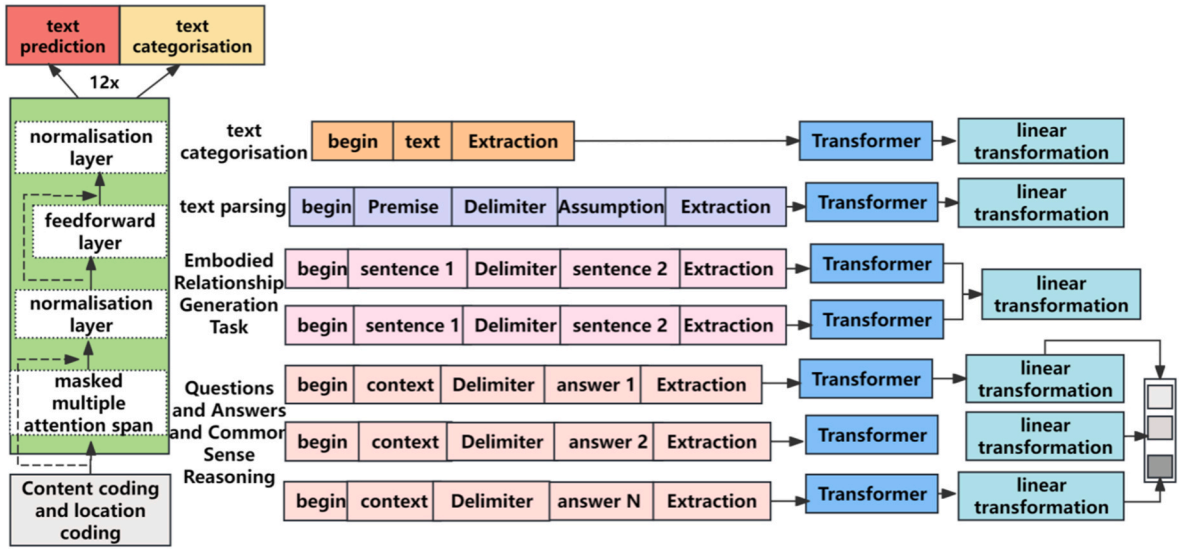


Fig. 2. Fine-tuning process for large language model, demonstrating diverse input formats and transformation into task-specific outputs for prediction and reasoning.

Table 1
Comparison of RAG and model fine-tuning.

Comparison term	RAG	Modelling fine-tuning
Definition	Combine large-scale knowledge base retrieval with large language model generation to retrieve relevant documents and use them to generate answers	Refinement of pre-trained models for specific tasks using smaller datasets, adjusting the weights of the models to make them more appropriate for specific tasks
Vantage	External knowledge can be used	It is possible to get good results on specific tasks, especially if you supervise sufficient data
Challenge	Need for a reliable recall system	There is a risk of over-fitting when there is insufficient data and the model can only handle scenarios before the training data cut-off date
Use case	Intelligent question and answer scenarios with frequently updated data	Using models for specific tasks, e. g. sentiment classification, intent recognition

results and presents it to the user. RAG is capable of generating high-quality answers, reducing illusions, having good scalability, and being able to trace information back to its source. Compared to fine-tuning, RAG requires only a certain amount of data to be used as an alternative method, reducing resource requirements.

The language model undergoes generative pre-training on various unlabeled text corpora, followed by discriminative fine-tuning for each specific downstream task.

2.3. LLM RAG for medicine

Retrieval augmented generation is emerging as a powerful technique for leveraging large language models in medicine. Liu et al. discuss the development principles of large language models in the medical field, their application performance in various medical tasks and clinical scenarios [26]. Chow et al. explored the potential applications and ethical challenges of large language models (LLMs) in healthcare, focusing on issues of accuracy, reliability, and bias in specific contexts (e.g., radiotherapy education chatbots), proposing mitigation strategies and frameworks for integrating emerging technologies to achieve safer and more effective medical decision-making and education [3,27–29]. Integrating task-specific calculation tools can improve the performance of large language models in medical calculation tasks, with LLaMa models reducing incorrect responses by 5.5-fold and GPT models by 13-fold [30]. Retrieval augmented generation outperforms domain specific fine-tuning in recall-oriented understudy for gisting evaluation, bilingual evaluation understudy, coverage score, with higher efficiency and fewer hallucinations [1]. Med-PaLM 2 performs excellently in medical question answering, approaching expert level with great application potential [14]. In summary, we present an overview of the literature, highlighting various implementations of RAG for medicine in Table 2.

3. Methods

The basic RAG can meet the demands of many tasks, but there are several issues, including imprecise retrieval due to semantic ambiguity leading to incorrect matches, improper context integration during the augmentation process, content redundancy, style inconsistencies, and such as hallucinations, misinformation, and length limitations during generation. We optimizes the base RAG model, using Qwen-14 B-chat as the foundational model, addressing these issues through improvements in data processing, prompt construction, and multi-retriever fusion. The overall model framework is shown in Fig. 3.

1) Data Processing

Data processing ensures that the data is clean, accurate, and contextually consistent, thereby improving the reliability of the RAG system. Text cleaning primarily involves normalizing text formats, removing special characters, and eliminating irrelevant information. Ambiguity between texts is resolved, for example, standardizing "wind-heat cold" and "heat cold" to a unified term, "wind-heat cold." Additionally, it involves removing duplicate content and redundant information from the text.

2) Prompt Construction

Prompt construction involves adding question answer examples in the input to guide the language model in understanding the task. For instance, a prompt template designed for medical knowledge reasoning tasks might be: "Please complete the natural language inference task. Given the premise and hypothesis, determine whether they have an entailment or no (or contradictory) relationship: Premise: {premise}, Hypothesis: {hypothesis}. If there is an entailment, meaning the premise

Table 2
A detailed list of LLM RAG for medicine.

Authors	Key points	LLMs	RAG method	Outcomes
Azimi et al. (2025) [2]	Registered Dietitian, Nutrition, Prompt Engineering, Knowledge Retrieval	GPT-4o, Claude 3.5 Sonnet, Gemini Pro 1.5	Chunking, Self-Supervised, Retrieval Augmented Prompting	Retrieval Augmented Prompting demonstrated exceptional efficacy in enabling GPT-4o to address expert-level questions
Das et al. (2025) [31]	Retrieval-augmented generation, substance use, social media	Nous-Hermes-2-7 B, GPT-4	Chunking, Retrieval Reasoning Summary	RAG effectively answers medical questions, even with limited resources
Duan et al. (2025) [32]	Large language model, knowledge graph, Traditional Chinese Medicine, question answering system	GPT-3.5-T, GLM-3-T, Qwen-T, Spark3.5 Max, Moonshot-V1-8K	Named Entity Recognition, knowledge, Fusion, few-shot learning	Large language models with knowledge graphs to enhance the efficiency and accuracy of Traditional Chinese Medicine case record retrieval
Fukushima et al. (2025) [33]	Large language models, genetic counseling, medical, retrieval-augmented generation, instruction tuning	JGCLLM	Self-Supervised, Enhanced prompt	RAG outperformed instruction tuning and prompt engineering in key areas of genetic counseling
Painter et al. (2025) [34]	Pharmacovigilance, SQL Generation, Contextual Query Optimization	GPT-4	Structured context, Indexing, Chunking, Self-Supervised	RAG framework enhances NL-to-SQL accuracy for pharmacovigilance queries
Woo et al. (2025) [35]	Medical information accuracy, Agentic Augmentation, Clinical Guidelines	GPT-4, GPT-3.5, Claude-3, LLaMA-3-8 B, LLaMA-3-70 B, Mixtral-8x7B	Agentic	RAG and agentic augmentation enhance medical question accuracy
Zhao et al. (2025) [36]	Healthcare Copilot, Retrieval-augmented Generation, Decision Support	Mixtral-8x7B, Qwen-2.5–72 B, LLaMA-3.1–70 B, LLaMA-3.1–8 B, GPT-3.5 T, GPT-4o-mini, GPT-4	Similarity Aggregate, Knowledge Graph elicited reasoning,	MedRAG with KG augmentation surpasses basic RAG
Zheng et al. (2025) [37]	Retrieval-augmented generation-based physician recommendation, triage systems	GPT-4o, GPT-4o-mini, Mistral-8x7B	Self-Supervised, Maximal Marginal Relevance	Knowledge-enhanced RAG model that improves accuracy and personalization in web-based medical consultations

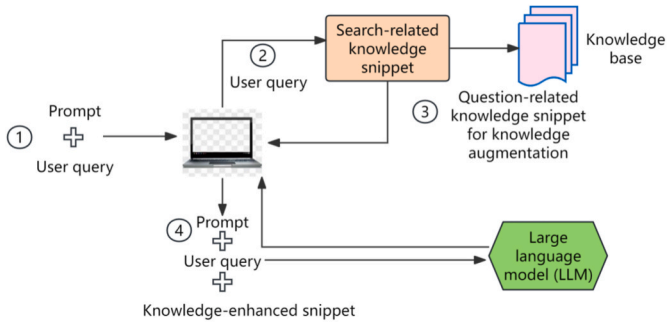


Fig. 3. The optimized RAG framework. This diagram details the process of input submission, knowledge retrieval, and integration with a large language model for enhanced output generation.

implies the hypothesis, or the hypothesis is likely the next sentence after the premise, or vice versa, output 0; otherwise, output 2." During debugging, the prompt includes 10 samples labeled as 0 and 2 from the training set to guide the model.

3) Multi-Retriever Fusion

The RAG improvement combines sparse and dense retrievers, leveraging their complementary strengths. Sparse retrievers excel at finding relevant texts based on keywords, while dense retrievers are better at retrieving based on semantic information. Both retrievers calculate the similarity score between the user query and candidate texts separately, and a fusion algorithm is then applied to compute a new similarity score. In this work, based on open-source medical language model corpora, a transformer is used as the dense retriever and TF-IDF as the sparse retriever, with an adaptive weight fusion method proposed to combine results. Term Frequency (TF) represents the frequency of a specific term t appearing in a document d.

$$TF(t,d) = \frac{f(t,d)}{\sum_{t^* \in d} f(t^*,d)} \tag{1}$$

$f(t,d)$ represents the number of times a word appears in a document, $\sum_{t^* \in d} f(t^*,d)$ denotes the total number of times all words appear in the document. Inverse Document Frequency (IDF) measures the importance of a word across the entire document collection, giving more weight to terms that appear in fewer documents, thus indicating their relevance.

$$IDF(t,D) = log\frac{N}{|\{d \in D : t \in d\}| + 1} \tag{2}$$

N represents the total number of documents in the collection, $|\{d \in D : t \in d\}|$ means the number of documents in which the term appears. Adding 1 helps to avoid division by zero. TF-IDF is the product of TF and IDF, as shown in equation (3).

$$\mathrm{TF-IDF(t,d,D)} = \mathrm{TF(t,d)}*\mathrm{IDF(t,D)} \tag{3}$$

The importance score of the word in the document relative to the whole document collection can be obtained. The dense searcher is a simplification and improvement of the original Transformer model, as shown in Fig. 4.

Transformer consists of an encoder and a decoder, the encoder consists of multiple identical layers stacked together. each layer includes a multi-head self-attention and a feed-forward neural network. Each sub-layer contains residual connectivity and layer normalization. Assuming the input sequence is $X = \{x_1, x_2, \ldots, x_n\}$, the input representation is shown in equation (4).

$$X^* = X + PE \tag{4}$$

where $x_i \in R^d$ is the feature representation and the positional encoding PE is calculated as shown in equation (5)-equation (6).

$$PE_{(pos,2i)} = sin\left(\frac{pos}{10000^{2i/d}}\right) \tag{5}$$

$$PE_{(pos,2i+1)} = sin\left(\frac{pos}{10000^{2i/d}}\right) \tag{6}$$

Where pos is the position index and i is the dimension index. For the input X^*, Multi-Head Self Attention mechanism computes the Query, Key and Value matrices as shown in equation (7).

$$\mathrm{Q} = \mathrm{X}^*\mathrm{W}^{\mathrm{Q}}, \mathrm{K} = \mathrm{X}^*\mathrm{W}^{\mathrm{K}}, \mathrm{V} = \mathrm{X}^*\mathrm{W}^{\mathrm{V}} \tag{7}$$

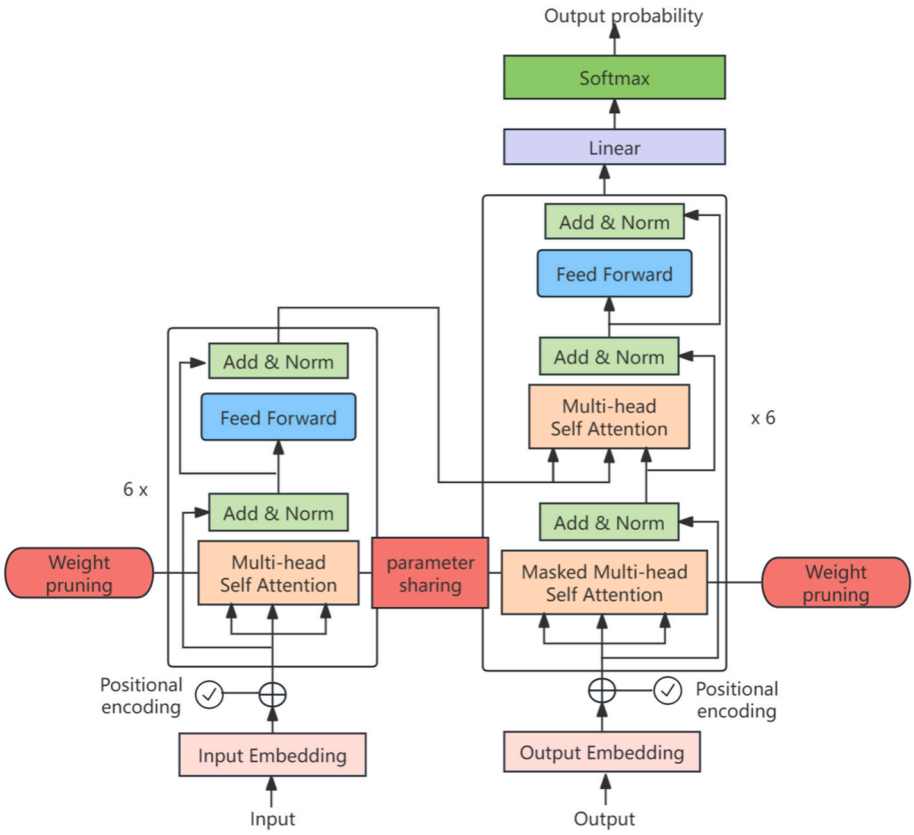


Fig. 4. Structure of the RAG dense retriever model, showing the transformer encoder and decoder blocks, input and output embedding with positional encoding, parameter sharing, and weight pruning.

where $W^Q, W^K, W^V \in R^{d*d_k}$ is the matrix of learnable parameters and d_k is the dimension of the attention head. Self-attention is calculated as shown in equation (8).

$$Attention(Q,K,V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{8}$$

multi-head attention computes attentions parallel, as shown in equation (9).

$$MultiHead(Q,K,V) = Concat(head_1,\ldots,head_h)W^O \tag{9}$$

where each attention head $head_i = Attention(Q_i,K_i,V_i)$, $W^O \in R^{hd_k*d}$ is a projection matrix. The output of the multi-head self-attention module is residual concatenated and layer normalized.

$$X^{(1)} = LayerNorm(X^* + MultiHead(Q,K,V)) \tag{10}$$

The Feed Forward Neural Network (FFNN) independently applies the same two-layer linear transformations and activation functions to the vector at each position.

$$\mathrm{FFN}\left(\mathrm{X}^{(1)}\right) = \mathrm{ReLU}\left(\mathrm{X}^{(1)}\mathrm{W}_1 + \mathrm{b}_1\right)\mathrm{W}_2 + \mathrm{b}_2 \tag{11}$$

where $W_1 \in R^{d*d_{ff}}$, $W_2 \in R^{d_{ff}*d}$, d_{ff} is usually greater than d. The feed forward network passes through residual connected and layer normalized. The output of each coding layer is used as the input of the next layer, and the whole encoder consists of a stack of N layers.

The decoder is composed of multiple identical layers stacked together, with each layer including a Masked Multi-Head Self-Attention module, an encoder-decoder attention module, and a feed-forward neural network. Each sub-layer contains residual connections and layer normalization. To prevent the decoder from seeing future information when generating the j_th word, the attention matrix is masked. The masked multi-head self-attention mechanism calculates the query, key, and value matrices as shown in equation (12).

$$Q_d = YW^{Q_d}, K_d = YW^{K_d}, V_d = YW^{V_d} \tag{12}$$

Masking is applied during the attention calculation, as shown in equation (13).

$$\mathrm{Attention}(\mathrm{Q_d},\mathrm{K_d},\mathrm{V_d}) = \mathrm{softmax}\left(\frac{\mathrm{Q_dK_d^T}}{\sqrt{\mathrm{d_k}}} + \mathrm{M}\right)\mathrm{V_d} \tag{13}$$

where the masking matrix M makes the location j only able to attend to the words of the location $<= j$. The multiple attention mechanism is as in equation (14).

$$\mathrm{MaskedMultiHead}(\mathrm{Q_d},\mathrm{K_d},\mathrm{V_d}) = \mathrm{Concat}(\mathrm{head_1},\ldots,\mathrm{head_h})\mathrm{W^{Q_d}} \tag{14}$$

The second attention layer of the decoder is used to focus on the encoder output, through which the query, key, and value matrices are computed to derive the multi-head attention score. The decoder output is used to generate the probability distribution of the next word through linear transformation and softmax. The encoder-decoder parameter sharing and multi-head attention random pruning operations are performed based on the traditional Transformer, which set $W^Q = W^{Q_d}$, $W^K = W^{K_d}$, $W^V = W^{V_d}$. The sparse retriever weight is set to W_{sparse} and the dense retriever weight is set to W_{dense}. make sure the sum of the two weights is 1. The similarity scores of TF-IDF and Transformer usually have different scales, so we use Min-Max normalization to normalize them to the interval [0, 1].

$$S_{sparse_norm} = \frac{S_{sparse} - min(S_{sparse})}{max(S_{sparse}) - min(S_{sparse})} \tag{15}$$

$$S_{dense_norm} = \frac{S_{dense} - min(S_{dense})}{max(S_{dense}) - min(S_{dense})} \quad (16)$$

where S_{sparse_norm} is the normalized sparse similarity score and S_{dense_norm} is the normalized dense similarity score. $min(S_{sparse})$ *and* $max(S_{sparse})$ are the minimum and maximum values of TF-IDF, and $min(S_{dense})$ and $max(S_{dense})$ are the minimum and maximum values of Transformer. The similarity scores of sparse and dense retrievers are fused by weighted summation as shown in equation (17).

$$S_{final} = W_{sparse} * S_{sparse_norm} + W_{dense} * S_{dense_norm} \quad (17)$$

W_{sparse} *and* W_{dense} are the weights of sparse and dense retrievers respectively, the sum of the weights is 1. Experiments were conducted using grid search for 10 sets of weights, and the optimal weights were $W_{sparse} = 0.35$, $W_{dense} = 0.65$.

4. Experiment

4.1. Data

Data comes from the CCKS-TCMBench Medical Knowledge Comprehension, the link is https://tianchi.aliyun.com/competition/entrance/532199/information. The assessment benchmark relies on the Medical Licensure Examination's rich question bank, which comprehensively covers the three major exam areas, including Basic Medical Theory, Clinical Medicine, and 16 core exam subjects, totaling 9788 questions and 5473 practice questions. The data includes 2 core tasks, (1) Medical knowledge comprehension assessment: based on the given questions, select the correct candidate answers and give the corresponding parses. (2) Medical semantic reasoning assessment: based on the given premises and assumptions, determine the relationship between them.

4.1.1. Medical knowledge understanding task

Firstly, the text-based exercises in PDF format are cleaned, and using custom rules, the questions, options, correct answers, subjects, and standard explanations are extracted. This information is then converted into a structured JSON format. It includes three types of questions, as shown in Table 3.

For A1/A2-type questions, the model's input is the content of the "question," and the output includes both the "answer" and the "analysis". For A3-type questions, each stem corresponds to a list of questions, consisting of multiple sub-questions. The model's input includes the shared stem, the sub-question, and the output of the previous sub-question. The target output is the "answer" and the "analysis" for each sub-question. For B1-type questions, each shared set of options corresponds to a list of questions, consisting of multiple sub-questions. The model's input includes the shared options, the sub-question, and the output of the previous sub-question. The target output is the "answer" and the "analysis" for each sub-question. To accommodate the different types of questions, separate instruction templates are constructed, which include a task description, a reasoning chain, and restrictions on the question-answer format. The task description clarifies how to approach each question type, while the reasoning chain is designed to guide the large language model in selecting an option and simultaneously providing the corresponding explanation.

Table 3
Composition of the three types of questions.

Question type	Data structure	Description
Single-stem Single Best Choice (A1/A2)	Question (stem) + options	Each question consists of one question (stem) with five options (A, B, C, D, E), where only one is the correct answer. There are two subtypes: A1-type and A2-type questions. A1-type: The stem is a statement, either affirmative or negative. A2-type: The stem presents a brief case summary.
Case Group Best Choice (A3)	Shared stem + question + options	The stem describes a patient-centered clinical scenario, followed by several related questions. Each question is linked to the same clinical scenario but tests different aspects. This type focuses on clinical application.
Standard Matching (B1)	Shared options + question (stem)	Each question consists of five options (A, B, C, D, E) followed by two or more questions (stems). The same set of five options applies to all the questions, and each option can be selected multiple times.

4.1.2. Medical semantic reasoning task

The data for Task 2 comes from the Tianchi Traditional Chinese Medicine AI Special Competition, and the data link is https://tianchi.aliyun.com/competition/entrance/532199/information. In the research, a total of 29,497 Natural Language Inference (NLI) data pairs were constructed. The hypothesis is that if there is stable semantic similarity between two sentences, it should be possible to infer the hypothesis from the premise and vice versa. Given the significant length differences between the problem answers and the standard explanations, half of the data pairs in NLI have undergone premise and hypothesis swapping to eliminate any bias caused by length. The model's input consists of the "premise" and "hypothesis," and the output is the "label". Thus, the task involves inputting the premise and hypothesis, and outputting the corresponding label. The sample data for Task 2 is as follows:

{ "idx": 0.

"premise": "Clinical infection, also known as symptomatic infection, refers to the invasion of pathogens into the human body, which not only triggers an immune response but also leads to tissue damage and pathological changes through the action of the pathogens themselves or the body's hypersensitivity reactions. Option A pertains to the clearance of pathogens. Option B pertains to asymptomatic infection. Option D pertains to latent infection. Option E pertains to the carrier state of pathogens.",
"hypothesis": "Medical care institutions must strictly implement disinfection and isolation protocols in accordance with the relevant regulations of the State Council's health administrative department to prevent hospital-acquired infections and iatrogenic infections.",
"label": 2

}

The label values are 0 and 2, where 0 indicates that there is an entailment relationship between the premise and the hypothesis, and 2 indicates that the premise and the hypothesis are unrelated or contradictory.

4.2. Evaluation metrics

There are two test sets for the two tasks, and we use separate scoring on each task for the evaluation. The evaluation metrics on each task are as follows.

1) **Medical knowledge Understanding task**
   a) We use accuracy to compare the correct options with the options provided by the model, in order to evaluate the model's mastery of medical knowledge. Let the standard answer be $T = \{t_1, t_2, ..., t_n\}$, and the model-generated answer be $P = \{p_1, p_2, ..., p_n\}$, then the accuracy is calculated as shown in formula (18).

$$Acc = \frac{1}{n} \sum_{i=1}^{n} I(t_i = p_i) \quad (18)$$

Where, $I(.)$ is the indicator function, when $t_i = p_i$, $I(t_i = p_i) = 1$, and 0 otherwise, n is the sample size.

b) By using Rouge-1, Rouge-L, BertScore, and SARI to automatically compare the semantic similarity between the explanations generated and the standard explanations, we can evaluate the model's ability to analyze medical knowledge.

Rouge-1: Calculates the overlap rate of 1-g between the explanations generated and the standard explanations, as shown in formula (19).

$$Rouge-1 = \frac{\sum_{i=1}^{n} count_{match-1-gram}(p_i, t_i)}{\sum_{i=1}^{n} count_{1-gram}(t_i)} \tag{19}$$

Rouge-L: Based on the longest common subsequence, it calculates the similarity between the explanations generated and the standard explanations, as shown in formula (20).

$$\text{Rouge} - \text{L} = \frac{\sum_{i=1}^{n} \text{LCS}(\text{p}_i, \text{t}_i)}{\text{length}(\text{t}_i)} \tag{20}$$

BertScore: Based on the BERT model, it calculates the embedding similarity between the explanations generated and the standard explanations, as shown in formula (21).

$$\text{BertScore} = \frac{1}{\text{n}} \sum_{\text{i}=1}^{\text{n}} \text{simBERT}(\text{p}_\text{i}, \text{t}_\text{i}) \tag{21}$$

SARI (System output Against References and Input): Evaluates the model's performance in the generated, inserted, and deleted parts, and is particularly suitable for text rewriting tasks.

$$\text{SARI} = \frac{1}{3}\left(Precision_{add} + Precision_{keep} + Precision_{delete}\right) \tag{22}$$

Precision measures the proportion of correct information generated by the system.

2) **Medical Semantic Reasoning Task**

Accuracy is used as the evaluation metric, and the calculation formula is the same as in the Knowledge Understanding Task.

3) **Overall Score**

The evaluation metrics for each task are averaged to calculate the overall score. Let the evaluation metrics for different tasks be $S_1, S_2, \ldots, S_m$, then the overall score S_{total} is calculated as shown in formula (23).

$$S_{total} = \frac{1}{m} \sum_{i=1}^{n} S_i \tag{23}$$

where m is the number of tasks, and S_i is the evaluation score for each task. Models are ranked based on the overall score and used as the evaluation.

4.3. Evaluation results

4.3.1. Model comparison

In this study, we use four models as baselines: Scholar Open-Source Large Language Model, ChatGLM3-6 B, ShenNong-TCM, and Qwen-14 B-chat. The results are shown in Table 4.

From Table 4 and it can be seen different large language model exhibit variations across multiple evaluation metrics. For instance, the accuracy of Qwen-14 B-chat is higher than that of other large language models, while ShenNong-TCM performs better on the RougeL metric. In the future, we will attempt to combine the strengths of different large language models to further enhance the results. Specifically.

- Qwen-14 B-Chat achieves the 0.9014 score on Acc-task2, indicating its strong generalization ability in classification tasks. It also performs well on Acc-task1 and SARI-task1, demonstrating its robustness in both semantic understanding and text generation.
- ShenNong-TCM scores 0.4521 on RougeL-task1, suggesting that its generated texts are more structurally aligned with the reference texts. This may be attributed to its specialized training in the field of traditional Chinese medicine, which enhances ability to generate coherent and well-structured content.
- ChatGLM3-6 B and Scholar show relatively lower performance across most metrics, particularly in SARI-task1 and Acc-task1, indicating potential limitations in text understanding and classification tasks.
- Our proposed method outperforms other models on all metrics, especially in SARI-task1 (0.3452) and Acc-task2 (0.9093), highlighting its superior performance in knowledge understanding and semantic reasoning.

From a practical perspective, these results indicate that different models have distinct strengths depending on the specific task. For example, if the task requires generating coherent and fluent text (e.g., summarization or dialogue generation), models with better Rouge scores, such as ShenNong-TCM, may be more suitable. In contrast, for classification tasks, Qwen-14 B-Chat or our method would be preferable.

The radar chart in Fig. 5 illustrates the performance of different models across multiple metrics. Our model stands out the most, leading other models in overall scoring, accuracy in Task 1 and Task 2, Rouge metrics, BertScore, and SARI metrics. In contrast, the Qwen-14 B-Chat closely follows, showing performance on multiple evaluation metrics that approaches ours. The Scholar large language model and ChatGLM3-6 B perform slightly less well. Overall, Our model demonstrates the strongest comprehensive ability when handling complex tasks.

4.3.2. Ablation experiments

The base model is Qwen-14 B-Chat, with improvements mainly in three areas: data processing, prompt construction, and RAG multi-retriever fusion. The ablation experiments separately test the effect of each improvement, and the results are shown in Table 5.

In Table 5, ABTest1 involves performing data processing on the Qwen-14 B-chat. ABTest2 includes prompt engineering optimization by adding representative positive and negative samples to the prompts. ABTest3 introduces a method of fusing dense and sparse retrievers within the basic RAG framework. It can be seen that each improvement brings enhancements, and when the three improvements are combined, the performance is further elevated. The percentage improvements compared to the base model are shown in Fig. 6.

Table 4
Indicators for the assessment of different models.

Model	Score	Acc-task1	Rouge1-task1	RougeL-task1	BertScore-task1	SARI-task1	Acc-task2
Scholar	0.5555	0.7948	0.4437	0.2073	0.6874	0.3120	0.8878
ChatGLM3-6 B	0.5536	0.7837	0.4525	0.2176	0.6852	0.3036	0.8791
ShenNong-TCM	0.5677	0.8144	0.4521	0.2336	0.7077	0.3128	0.8857
Qwen-14 B-Chat	0.5670	0.8215	0.4511	0.2236	0.6902	0.3143	0.9014
Ours	**0.5826**	**0.8316**	**0.4635**	**0.2346**	**0.7112**	**0.3452**	**0.9093**

4.3.3. Realistic prediction results

Task 1 Real Case:


```
{
    "keywords": "test_A12",
    "model_name": "Ours LLM based Qwen-14B-chat",
    "prompt": "Please answer an A1 or A2 type multiple-choice question from a medical test. Please think step by step and write your thought process between【Explanation】and <eoe>. You will select the most correct answer from A, B, C, D, E, and write it between【Answer】and <eoa>. For example:【Answer】: A <eoa> , The complete format of the answer is as follows: 【Explanation】 ... <eoe>,【Answer】... <eoa>, Please strictly follow the above format when answering. The question is as follows:",
    "example": [
        {
            "index": 0,
            "question": "To treat shoulder pain, especially when the posterior shoulder is heavily affected and the pain resists pressure, in addition to shoulder acupoints, which main meridian points should be selected? ( ) A. Hand Taiyang Small Intestine Meridian B. Hand Yangming Large Intestine Meridian C. Hand Shaoyang Triple Burner Meridian D. Foot Shaoyang Gallbladder Meridian E. Foot Taiyang Bladder Meridian",
            "answer": [
                "A"
            ],
            "analysis": "【analysis】Shoulder pain, especially when the posterior shoulder is heavily affected and the pain resists pressure, is usually related to the meridians and muscle tissues of the shoulder. Shoulder pain is generally closely associated with the Hand Shaoyang Triple Burner Meridian and the Hand Taiyang Small Intestine Meridian because the Hand Shaoyang Triple Burner Meridian extends from the fingertips to the shoulder, and the Hand Taiyang Small Intestine Meridian mainly passes through the shoulder and neck. Based on the characteristic of pain resisting pressure, it may involve stagnation of qi and blood in the meridians or local inflammation. Therefore, choosing the Hand Taiyang Small Intestine Meridian is more appropriate because the qi of the Small Intestine Meridian often manifests as pain that resists pressure when affected by illness.【answer】 A <eoa>"
        },
}
```


Task 2 Real Case:


```
{
    "keywords": "NLI task",
    "example": [
        {
            "idx": 0,
            "premise": "Among the five viscera, the liver is the one that has the characteristic of a "hard organ."",
            "hypothesis": "The lung is considered a delicate organ, the heart is the fire organ, the spleen is the earth organ, the liver is the hard organ, and the kidney is the organ of water and fire.",
            "label": 0
        },
}
```

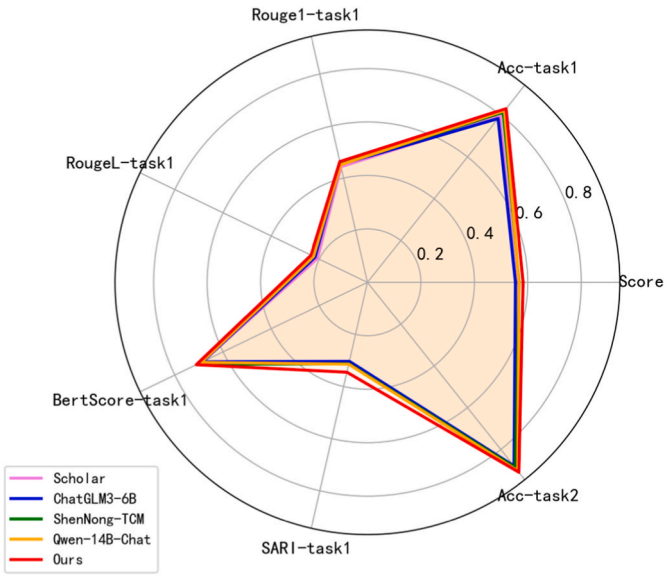


Fig. 5. Radar evaluation of different large language models across multiple performance metrics.

5. Conclusion

Based on over 15,000 sample data entries from the Medical Knowledge Understanding and Reasoning Ability Evaluation Competition, our research focuses on the tasks of medical knowledge understanding and medical semantic reasoning. Using Qwen-14 B-Chat as the base model, we propose an improved retrieval-augmented generative modeling algorithm. The main improvements include data processing, prompt selection, and the fusion of dense and sparse retrievers. Compared to four large base models, the average comprehensive evaluation metrics have increased by +3.86 %, verifying the effectiveness of our method. Additionally, ablation experiments demonstrate that each improvement enhances the base model.

However, this study still has limitations: firstly, the data originates from a competition, and its domain coverage may not fully represent the real-world clinical diversity. secondly, evaluation mainly relies on standard metrics, lacking in-depth exploration of the model's performance in actual clinical environments. In the future, we plan to explore several key directions: expanding data sources by incorporating diverse real clinical case to enhance the model's generalization and understanding of rare diseases. Developing more explainable versions to improve transparency and foster trust among doctors. Integrating the method into clinical decision support systems and intelligent consultation platforms to optimize the model through practical feedback, and investigating its potential value in areas like medical education and auxiliary diagnosis.

CRediT authorship contribution statement

Yingshuai Wang: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Yanli Wan:** Writing – review & editing, Visualization, Project administration. **Xingyun Lei:** Writing – review & editing, Software. **Qingkun Chen:** Writing – review & editing, Investigation. **Hongpu Hu:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Patient and public involvement

It was not appropriate or possible to involve patients or the public in the design, or conduct, or reporting, or dissemination plans of our research.

Table 5
Results of ablation experiments.

experiments	Score	Acc-task1	Rouge1-task1	RougeL-task1	BertScore-task1	SARI-task1	Acc-task2
Base	0.5670	0.8215	0.4511	0.2236	0.6902	0.3143	0.9014
ABTest1	0.5712	0.8276	0.4521	0.2306	0.6935	0.3178	0.9058
ABTest2	0.5722	0.8194	0.4503	0.2295	0.7024	0.3345	0.8972
ABTest3	0.5741	0.8281	0.4526	0.2288	0.7062	0.3269	0.9021
Ours	**0.5826**	**0.8316**	**0.4635**	**0.2346**	**0.7112**	**0.3452**	**0.9093**

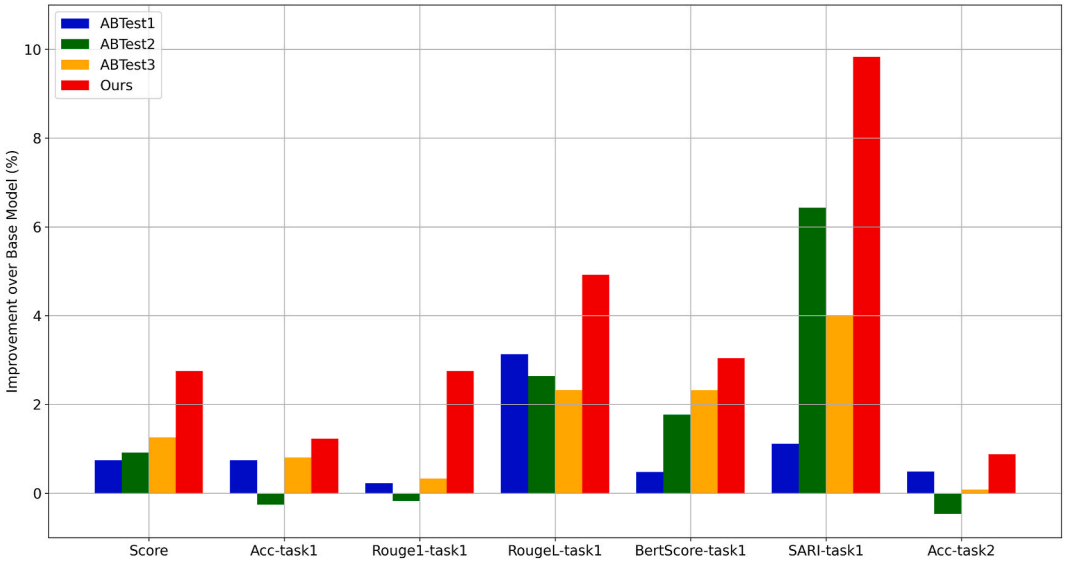


Fig. 6. Percentage improvement of ablation experiments over baseline models across different tasks.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] Lakatos R, Pollner P, Hajdu A, et al. Investigating the performance of retrieval-augmented generation and domain-specific fine-tuning for the development of AI-driven knowledge-based systems. Machine Learning and Knowledge Extraction 2025;7(1):15.
[2] Azimi I, Qi M, Wang L, et al. Evaluation of LLMs accuracy and consistency in the registered dietitian exam through prompt engineering and knowledge retrieval. Sci Rep 2025;15(1):1506.
[3] Chow JCL, Li K. Ethical considerations in human-centered AI: advancing oncology chatbots through large language models. JMIR Bioinformatics and Biotechnology 2024;5:e64406.
[4] Luo MJ, Pang J, Bi S, et al. Development and evaluation of a retrieval-augmented Large Language Model framework for ophthalmology. JAMA ophthalmology 2024.
[5] Xiong G, Jin Q, Lu Z, et al. Benchmarking retrieval-augmented generation for medicine. arXiv preprint arXiv:240213178 2024.
[6] Long C, Liu Y, Ouyang C, et al. Bailicai: a domain-optimized retrieval-augmented generation framework for medical applications. arXiv preprint arXiv:240721055 2024.
[7] Kresevic S, Giuffrè M, Ajcevic M, et al. Optimization of hepatological clinical guidelines interpretation by large language models: a retrieval augmented generation-based framework. npj Digit Med 2024;7(1):102.
[8] Lozano A, Fleming SL, Chiang CC, et al. Clinfo. ai: an open-source retrieval-augmented large language model system for answering medical questions using scientific literature[C]. Pac Symp On Biocomputing 2024 2023:8–23.
[9] Jeong M, Sohn J, Sung M, et al. Improving medical reasoning through retrieval and self-reflection with retrieval-augmented large language models. Bioinformatics 2024;40:119–29.
[10] Amugongo LM, Mascheroni P, Brooks SG, et al. Retrieval augmented generation for Large Language models in healthcare. Syst Rev 2024.
[11] Salemi A, Zamani H. Evaluating retrieval quality in retrieval-augmented generation[C]. In: Proceedings of the 47th international ACM SIGIR conference on research and development in information retrieval; 2024. p. 2395–400.
[12] Chen J, Lin H, Han X, et al. Benchmarking large language models in retrieval-augmented generation[C]. Proc AAAI Conf Artif Intell 2024;38(16):17754–62.
[13] Es S, James J, Anke LE, et al. RAGAs: automated evaluation of retrieval augmented generation[C]. In: Proceedings of the 18th conference of the European chapter of the association for computational linguistics: system demonstrations; 2024. p. 150–8.
[14] Singhal K, Tu T, Gottweis J, et al. Toward expert-level medical question answering with large language models. Nat Med 2025;31(3):943–50.
[15] Chang Y, Wang X, Wang J, et al. A survey on evaluation of large language models. ACM Transactions on Intelligent Systems and Technology 2024;15(3):1–45.
[16] Tu W, Deng W, Gedeon T. A closer look at the robustness of contrastive language-image pre-training (clip). Adv Neural Inf Process Syst 2024;36.
[17] Nievas M, Basu A, Wang Y, et al. Distilling large language models for matching patients to clinical trials. J Am Med Inf Assoc 2024:ocae073.
[18] Zhang D, Chen L, Zhang S, et al. Large language models are semi-parametric reinforcement learning agents. Adv Neural Inf Process Syst 2024:36.
[19] Cascella M, Semeraro F, Montomoli J, et al. The breakthrough of large language models release for medical applications: 1-year timeline and perspectives. J Med Syst 2024;48(1):22.
[20] Biswas SS. Role of chat gpt in public health. Ann Biomed Eng 2023;51(5):868–9.
[21] Singhal K, Azizi S, Tu T, et al. Large language models encode clinical knowledge. Nature 2023;620(7972):172–80.
[22] Strickland E, Watson IBM, thyself heal. How IBM overpromised and underdelivered on AI health care. IEEE Spectrum 2019;56(4):24–31.
[23] Ramgopal S, Sanchez-Pinto LN, Horvat CM, et al. Artificial intelligence-based clinical decision support in pediatrics. Pediatr Res 2023;93(2):334–41.
[24] Lewis P, Perez E, Piktus A, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. Adv Neural Inf Process Syst 2020;33:9459–74.
[25] Zhu K, Zhao Q, Chen H, et al. Promptbench: a unified library for evaluation of large language models. J Mach Learn Res 2024;25(254):1–22.
[26] Liu F, Zhou H, Gu B, et al. Application of large language models in medicine. Nature Reviews Bioengineering 2025:1–20.
[27] Chow JCL. Quantum computing and machine learning in medical decision-making: a comprehensive review. Algorithms 2025;18(3):156.
[28] Chow JCL, Li K. Developing effective frameworks for large language model-based medical chatbots: insights from radiotherapy education with ChatGPT. JMIR cancer 2025;11(1):e66633.
[29] Chow JCL, Sanders L, Li K. Impact of ChatGPT on medical chatbots as a disruptive technology. Frontiers in artificial Intelligence 2023;6:1166014.
[30] Goodell AJ, Chu SN, Rouholiman D, et al. Large language model agents can use tools to perform clinical calculations. npj Digit Med 2025;8(1):163.
[31] Das S, Ge Y, Guo Y, et al. Two-layer retrieval-augmented generation framework for low-resource medical question answering using reddit data: proof-of-concept study. J Med Internet Res 2025;27:e66220.
[32] Duan Y, Zhou Q, Li Y, et al. Research on a traditional Chinese medicine case-based question-answering system integrating large language models and knowledge graphs. Front Med 2025;11:1512329.
[33] Fukushima T, Manabe M, Yada S, et al. Evaluating and enhancing Japanese large language models for genetic counseling support: comparative study of domain adaptation and the development of an expert-evaluated dataset. JMIR Med Inform 2025;13:e65047.
[34] Painter JL, Chalamalasetti VR, Kassekert R, et al. Automating pharmacovigilance evidence generation: using large language models to produce context-aware structured query language. JAMIA open 2024;8(1).
[35] Woo JJ, Yang AJ, Olsen RJ, et al. Custom large language models improve accuracy: comparing retrieval augmented generation and artificial intelligence agents to noncustom models for evidence-based medicine. Arthrosc J Arthrosc Relat Surg 2025;41(3):565–573. e6.
[36] Zhao X, Liu S, Yang SY, et al. Medrag: enhancing retrieval-augmented generation with knowledge graph-elicited reasoning for healthcare copilot[C]. In: Proceedings of the ACM on web conference 2025; 2025. p. 4442–57.
[37] Zheng Y, Yan Y, Chen S, et al. Integrating retrieval-augmented generation for enhanced personalized physician recommendations in web-based medical services: model development study. Front Public Health 2025;13:1501408.